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Technological relatedness, boundary-spanning combination of knowledge and the impact of innovation: Evidence of an inverted-U relationship

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ABSTRACT

Our paper proposes that corporate technological relatedness, or the degree to which business units within a corporation utilize similar technological knowledge, has both positive and negative effects on corporate R&D activities. On the one hand, business units that employ similar technological knowledge have better absorptive capacity to source knowledge from each other. On the other hand, a higher level of technological relatedness means that each business unit possesses fewer opportunities to gain new knowledge not known to other units, thus promoting path dependence to each other. Using a patent data analysis of 201 firms in R&D-intensive industries, we examine the effects of corporate technological relatedness on within-firm knowledge flow, boundary-spanning combinations of prior knowledge, and innovation impacts.

1. Introduction

In strategic management theory, the resource-based view suggests that firms should avoid operating business portfolios that require vastly different kinds of resources and capabilities. According to this view, related diversifiers that exploit identical resources and capabilities across different businesses perform better than unrelated diversifiers unable to create synergistic effects. Thus, this view recommends that firms develop core competences to excel in a small number of aspects along value chain activities and join industries where they can exploit those competences. This theoretical argument has generated many empirical studies that vary with respect to the kinds of resources examined and the methodological approaches taken to measure relatedness among different business units.

Recent studies have suggested that achieving economies of scope based on technological knowledge is one of the most appropriate reasons for diversification (Miller, 2004; Robins & Wiersema, 1995). The costs associated with additional application of technological knowledge within a corporate boundary are trivial. At the same time, transferring knowledge through a market mechanism is cumbersome, since devising contracts sophisticated enough to prevent opportunistic behavior by both knowledge seller and buyer is difficult. In line with this argument, recent empirical studies have demonstrated that firms creating synergy based on technological knowledge exhibit superior performance than those do not (Miller, 2006; Silverman, 1999; Tanriverdi & Venkatraman, 2005). These studies imply that corporation with business units of similar technological expertise will perform better than those with business units that pursue different technological paths.

Although the similarity of technological expertise among a corporation's subunits may create synergy in R&D activities, excessive technological overlap may undermine a firm's R&D performance. Within the context of overseas R&D networks of multinational corporations, Song and Shin (2008) argued that R&D units have little incentive to source knowledge from each other when they have similar technological capabilities, thereby learning little from each other. On the other hand, innovations

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